Cognitive Modeling

Complete Lecture Notes

Week 14: Modeling in Cognitive Science I: Categorization (GCM) and Decision Making (Drift-Diffusion)

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Chapter 1

Applications in Cognitive Science

1.1 Week 14: Modeling in Cognitive Science I: Categorization (GCM) and Decision Making (Drift-Diffusion)

> Chapter 6: Examples of modeling in different areas of cognitive science · Lecture 1 h · Lab 2 h Running data: simulated classification data for the classic 5-4 category structure, and simulated choice-RT data from a drift-diffusion process in a speed/accuracy × easy/hard design.

1.1.1 Key terms

Important terms of this lesson, with their Persian equivalents (repeated even if they appeared in earlier weeks).

English	فارسی	توضیح کوتاه
categorization	دسته‌بندی	قرار دادن محرک در یک دسته
Generalized Context Model (GCM)	مدل بافت تعمیم‌یافته	مدل exemplar دسته‌بندی بر پایه‌ی شباهت
drift rate	نرخ رانش	سرعت میانگین جمع شدن شواهد؛ نشانه‌ی کیفیت شواهد
simulation	شبیه‌سازی	تولید داده‌ی مصنوعی با اجرای مدل
response time (RT)	زمان واکنش	فاصله‌ی ارائه‌ی محرک تا پاسخ
drift-diffusion model (DDM)	مدل drift-diffusion (رانش–انتشار)	انباشت پیوسته‌ی شواهد بین دو مرز
speed–accuracy trade-off	بده‌بستان سرعت و دقت	سریع‌تر پاسخ دادن به قیمت خطای بیشتر
accuracy	دقت	نسبت پاسخ‌های درست

English	فارسی	توضیح کوتاه
exemplar	نمونه (exemplar)	یک مورد ذخیره‌شده در حافظه از یک دسته
prototype	پیش‌نمونه (prototype)	خلاصه یا میانگین یک دسته
fit / fitting	برازش	تنظیم پارامترها تا مدل با داده جور شود
attention weight	وزن توجه	سهم توجه به هر بُعد محرک
sensitivity (d′)	حساسیت	فاصله‌ی توزیع علامت و نویز
parameter	پارامتر	کمیت قابل تنظیم مدل که بین افراد یا شرایط فرق می‌کند
estimate / estimator	برآوردگر / برآورد	روش به / مقدار به‌دست‌آمده برای پارامتر دست آوردن آن
EZ-diffusion	EZ-diffusion	برآورد ساده‌ی پارامترهای از سه DDM آماره
parameter recovery	بازیابی پارامتر	آیا برازش، پارامترهای معلوم شبیه‌سازی را برمی‌گرداند
evidence	شواهد	اطلاعاتی که برای تصمیم جمع می‌شود
stimulus	محرک	چیزی که به شرکت‌کننده نشان داده می‌شود
boundary separation	فاصله‌ی مرزها	مقدار شواهد لازم برای پاسخ؛ نشانه‌ی احتیاط
non-decision time	زمان غیرتصمیمی	زمان دیدن محرک و اجرای حرکت پاسخ
similarity	شباهت	نزدیکی دو محرک در فضای روان‌شناختی
trial	کوشش (trial)	یک بار انجام تکلیف توسط شرکت‌کننده
embedding	embedding (بردار معنایی)	برداری که مدل برای یک کلمه یا متن می‌سازد

1.1.2 Learning outcomes

By the end of this week, students will be able to:

1. Describe exemplar and prototype theories of categorization and formalize them as the GCM and a prototype model.
2. Fit the GCM and the prototype model to classification data by MLE, compare them, and interpret attention weights and sensitivity.
3. Describe the drift-diffusion model (DDM) as the continuous limit of the Week 1 random walk, and map its parameters onto cognitive constructs.
4. Estimate DDM parameters with the closed-form EZ-diffusion method and validate it by parameter recovery.

5. Interpret experimental effects in terms of model parameters ("this manipulation changed caution, not evidence quality").

1.1.3 Motivation & big picture

The first 13 weeks built a toolkit, mostly on two small running examples (forgetting and SDT). Chapter 6 shows the toolkit at work on models that have shaped whole research areas. This week covers two of the most influential.

Categorization. How do we decide that a new animal is a dog? *Prototype theory* says that we compare it with a summary of the category, the average dog. *Exemplar theory* says that we compare it with stored memories of every dog (and every cat) we have seen. Both sound plausible, and verbal arguments went back and forth for years. Formal models made the difference: Medin and Schaffer's 5-4 category structure was designed so that the two theories make opposite predictions for specific stimuli. Nosofsky's Generalized Context Model (GCM) turned exemplar theory into a precise, fit-able model with interpretable attention weights.

Decision making. In Week 1 we built a random walk that accumulates evidence to a boundary. Its continuous version, the drift-diffusion model, is arguably the most successful model in cognitive psychology: it explains accuracy *and* the full RT distributions of correct and error responses, with parameters for evidence quality, caution and non-decision time. It is used to study ageing, ADHD, depression, learning and the effect of drugs. Fitting the full DDM is computationally heavy, but the EZ-diffusion method gives good estimates from three summary statistics with closed-form equations.

Where is this used?

- Categorization models explain how people learn concepts (medical students learning to recognize skin lesions, children learning words), and they are used to design better training: which examples to show, and in what order.
- Exemplar ideas in machine learning: nearest-neighbor classifiers and kernel methods classify a new case by its similarity to stored examples, the same principle as the GCM.
- The drift-diffusion model is one of the most used models in cognitive neuroscience and clinical research: ageing, ADHD, depression, anxiety, effects of drugs and sleep deprivation. It is also used in consumer research (how fast people choose between products) and in neuroeconomics.
- EZ-diffusion is popular because it needs only three summary statistics per person and no optimization, so it can be applied to existing datasets quickly.

1.1.4 Lecture notes

In plain words Two famous models this week. Categorization: how do we decide that a new animal is a dog? One theory says we compare it with an average "ideal dog" (prototype). Another says we compare it with all the dogs and cats we remember and see which group it resembles more overall (exemplars). The GCM turns the exemplar idea into math: measure the distance between the new item and each stored example (paying more attention to some features than others), turn distance into similarity (closer = much more similar), and choose category A with probability "total similarity to A / total similarity to all". A clever set of stimuli (the 5-4 structure) makes the two theories predict *opposite* things for some items. Decisions: the drift-diffusion model is the continuous version of Week 1's random walk. Evidence drifts toward the correct boundary at an average speed v (quality of evidence) while being jiggled by noise; a response is made when a boundary a (caution) is reached; a fixed time T_{er} is added for seeing and pressing. EZ-diffusion is a shortcut that computes v, a and T_{er} from just three numbers: accuracy, and the mean and variance of correct RTs.

Suggested timing (60 min): categorization theories and the 5-4 structure 10 · GCM 15 · prototype model and the critical predictions 5 · the drift-diffusion model 15 · EZ-diffusion 10 · interpretation 5.

1. The 5-4 category structure

Stimuli vary on four binary dimensions (e.g., the shape, color, size and position of a geometric figure in Medin & Schaffer's design; facial features in later versions). Participants learn to classify 9 training stimuli into categories A and B, then classify 7 new transfer stimuli without feedback. The commonly used stimulus set is [verify against Medin & Schaffer, 1978, or Smith & Minda, 2000]:

Category A	Category B	Transfer
A1 1110	B1 1100	T1 1001
A2 1010	B2 0110	T2 1000
A3 1011	B3 0001	T3 1111
A4 1101	B4 0000	T4 0010
A5 0111		T5 0101
		T6 0011
		T7 0100

The category A prototype is 1111 and the B prototype is 0000. The critical comparison: A1 (1110) shares three features with the A prototype, while A2 (1010) shares only two. A prototype model therefore predicts that A1 is classified *better* than A2. But A2 is highly similar to two other A exemplars (A1 and A3) and to no B exemplar, whereas A1 is highly similar to two B exemplars (B1 and B2). An exemplar model can therefore predict that A2 is classified *at least as well* as A1, and in the empirical data A2 is typically classified better.

2. The Generalized Context Model (GCM)

Distance between stimulus i and stored exemplar e, with attention weights $w_m \geq 0$, $\sum_m w_m = 1$ (city-block metric for separable dimensions):

$$\text{dist}_{ie} = \sum_{m=1}^{4} w_m \, |x_{im} - x_{em}|.$$

Similarity decays exponentially with distance, with sensitivity c:

$$s_{ie} = \exp(-c \cdot \text{dist}_{ie}).$$

Choice rule (summed similarity, Luce ratio):

$$P(A \mid i) = \frac{\sum_{e \in A} s_{ie}}{\sum_{e \in A} s_{ie} + \sum_{e \in B} s_{ie}}.$$

Parameters: c (how steeply similarity falls with distance; large c means only very close exemplars matter) and three free attention weights (the fourth is \$1 - \$ the others), so $k = 4$. (This is the overloaded c from the Week 1 notation table; this week it means sensitivity, not an SDT criterion.)

Interpretation of attention weights: w_m is the share of attention given to dimension m. Optimal attention emphasizes dimensions that are diagnostic of the category. In the 5-4 structure, dimensions 1 and 3 are the most diagnostic.

Prototype model: the same distance, similarity and choice rule, but each category is represented by its prototype only: $P(A \mid i) = s_{i,P_A}/(s_{i,P_A} + s_{i,P_B})$. Also $k = 4$, so the comparison is not affected by parameter counting.

Likelihood: if stimulus i is classified n_i times and called "A" y_i times, then $y_i \sim \text{Binomial}(n_i, P(A \mid i))$, fitted by MLE with multi-start (Weeks 3–4). Weights are best estimated through a softmax reparameterization (Week 3), $w = \text{softmax}(\eta_1, \eta_2, \eta_3, 0)$, and $c = e^{\gamma}$.

> Advanced: the full GCM adds a response-bias parameter b_A, a similarity exponent (Gaussian vs. exponential similarity), a metric parameter r (city-block $r = 1$ vs. Euclidean $r = 2$) and sometimes a response-determinism parameter γ (raising summed similarities to a power). Lee & Wagenmakers (chapter 17) develop a Bayesian, hierarchical GCM.

Reading the math

- Distance $\sum_m w_m |x_{im} - x_{em}|$: count the features on which the new item i and the stored example e differ, but weight each feature by its attention w_m. With 4 binary features and equal weights 0.25, one mismatching feature gives distance 0.25.
- Similarity $e^{-c \cdot \text{dist}}$: distance 0 gives similarity 1; with $c = 2$, distance 0.25 gives $e^{-0.5} = 0.61$, distance 1 gives $e^{-2} = 0.14$. A larger c makes similarity drop faster, so only very close examples matter.

- Choice rule: add the similarities to all A examples and to all B examples, and divide. In the worked example, T3 has total similarity 2.79 to A and 1.09 to B, so $P(A) = 2.79/3.89 = 0.72$.

3. The drift-diffusion model (DDM)

Take the Week 1 random walk and let the step size and step duration shrink together. Evidence $X(t)$ then evolves continuously:

$$dX = v\,dt + s\,dW(t), \qquad X(0) = z,$$

where dW is Gaussian white noise (a Wiener process). A response is made when X reaches a (correct) or 0 (error); RT $= T_{\text{decision}} + T_{er}$.

Parameter	Meaning	Typical manipulation that changes it
v (drift rate)	Quality / rate of evidence accumulation	Stimulus difficulty; perceptual sensitivity; attention
a (boundary separation)	Response caution (speed–accuracy setting)	Speed vs. accuracy instructions
z (starting point)	A priori bias toward one response	Base rates, payoffs
T_{er} (non-decision time; also written t_0)	Encoding + motor time	Response modality, stimulus degradation
s (within-trial noise)	Scaling constant	Fixed (by convention $s = 0.1$ in EZ, $s = 1$ elsewhere)

The full DDM also includes across-trial variability in drift, starting point and non-decision time. Week 1, Exercise 5 showed that drift variability produces slow errors.

Signature predictions: right-skewed RT distributions; a speed–accuracy trade-off when a changes; easier stimuli giving responses that are faster *and* more accurate (higher v); a change in T_{er} shifting the whole RT distribution without changing its shape.

Reading the math $dX = v\,dt + s\,dW(t)$ reads: "in a tiny time step dt, the evidence changes by a deterministic push $v\,dt$ plus a random kick $s\,dW$". dW is Gaussian noise whose variance is proportional to dt (so over a step of length dt the kick has SD $s\sqrt{dt}$). In the lab simulation this becomes exactly Week 1's random walk with small steps: `x+=v*dt+s*sqrt(dt)*randn()`. The table of parameters is the important part to memorize: v = evidence quality, a = caution, z = bias, T_{er} = non-decision time.

4. EZ-diffusion

Wagenmakers, van der Maas and Grasman (2007) derived closed-form estimates of v, a and T_{er} (assuming $z = a/2$, no across-trial variability, $s = 0.1$) from three statistics: the proportion correct P_c, the variance of correct RTs V_{RT} (in s²) and the mean of correct RTs M_{RT} (in s):

$$L = \text{logit}(P_c) = \ln \frac{P_c}{1 - P_c}, \qquad x = \frac{L\,(LP_c^2 - LP_c + P_c - 0.5)}{V_{RT}},$$

$$v = \text{sign}(P_c - 0.5)\; s\; x^{1/4}, \qquad a = \frac{s^2 L}{v},$$

$$y = -\frac{va}{s^2}, \qquad \text{MDT} = \frac{a}{2v} \cdot \frac{1 - e^y}{1 + e^y}, \qquad T_{er} = M_{RT} - \text{MDT}.$$

Edge cases: the formulas fail at $P_c = 0.5$ (then $v = 0$) and $P_c = 1$ (then $L = \infty$). For $P_c = 1$, a standard fix is to use $P_c = 1 - 1/(2N)$, analogous to the extreme-rate corrections of Week 2.

Strengths and limits. EZ is fast, needs no optimization and recovers parameters well when its assumptions hold (lab Part C). It ignores error RTs and the shape of the RT distribution beyond mean and variance, and it cannot estimate bias or across-trial variabilities. For those, fit the full DDM by MLE or with hierarchical Bayesian methods (e.g., the HDDM package or the Wiener likelihood in PyMC/Stan).

Reading the math The EZ equations look intimidating but are a recipe; nobody needs to memorize them.

1. $L = \ln(P_c/(1 - P_c))$: the logit of accuracy ("log odds of being correct"). $P_c = 0.8$ gives $L = \ln 4 = 1.39$.
2. x combines accuracy with the variance of correct RTs: the more variable the RTs for a given accuracy, the lower the drift.
3. v and a then follow from L and x; MDT is the predicted *mean decision time*.
4. $T_{er} = M_{RT} - \text{MDT}$: whatever part of the mean RT is not decision time is non-decision time.

Use the worked example (Part 2) to show each step with numbers; then let the computer do it (the function `ez_diffusion` in the lab code).

For an engineer

- GCM is a similarity-based classifier with an exponential kernel and a weighted L1 (city-block) metric, close to a kernel / Parzen-window classifier. The attention weights act like learned feature weights; c is the inverse kernel width.
- The DDM is a Wiener process with drift between two absorbing boundaries: the classic first-passage-time problem. The continuous-time SPRT (Week 1 box) is exactly this process, with $v \propto$ signal amplitude and s the noise level.
- RT distributions are first-passage-time distributions, which are right-skewed for the same reason detection delays of a sequential detector are.

- EZ-diffusion is a method-of-moments estimator: it matches three sample moments (accuracy, mean and variance of correct RTs) to their closed-form model expressions, instead of maximizing a likelihood.

For the instructor

- Let students be the participants. Before the GCM, show the 9 training stimuli of the 5-4 structure as simple drawings (e.g. shapes with 4 binary features), teach the categories with feedback, then test A1 and A2. The class's own data often show the exemplar pattern.
- Draw the DDM as Week 1's picture with a smooth, noisy path. Then change one parameter at a time on the board (wider boundaries, steeper drift, later start) and ask what happens to accuracy and RT.
- Do not derive EZ. Present it as a "calculator" and spend the time on interpretation (lab Part E): which manipulation changed which parameter.
- Typical questions.
  - *"Prototype or exemplar, who won?"* For small, poorly structured categories exemplar models usually fit better; for large, well-structured ones prototypes do well. Both ideas survive in modern models.
  - *"Why not just compare mean RTs?"* A slower group could be more cautious, have worse evidence, or be slower motorically; only the DDM separates them.
  - *"Are the parameters of EZ and the full DDM the same?"* They estimate the same quantities, but EZ assumes no bias and no trial-to-trial variability.

1.1.5 Worked example

Part 1: one GCM prediction by hand

Stimulus T3 (1111), equal attention $w_m = 0.25$, sensitivity $c = 2$.

Exemplar	Mismatching dimensions	Distance	Similarity $e^{-2\,\text{dist}}$
A1 1110	1	0.25	0.607
A2 1010	2	0.50	0.368
A3 1011	1	0.25	0.607
A4 1101	1	0.25	0.607
A5 0111	1	0.25	0.607
Sum A			2.794
B1 1100	2	0.50	0.368
B2 0110	2	0.50	0.368

Exemplar	Mismatching dimensions	Distance	Similarity $e^{-2\,\text{dist}}$
B3 0001	3	0.75	0.223
B4 0000	4	1.00	0.135
Sum B			1.094

$$P(A \mid \text{T3}) = \frac{2.794}{2.794 + 1.094} = 0.719.$$

Interpretation. T3 is the A prototype, yet the exemplar model predicts only about 72% "A" responses with these parameters: two B exemplars (B1 and B2) are only two features away. A larger c (sharper similarity) would raise this probability, because the nearest neighbors (four A exemplars at distance 0.25) would dominate.

Part 2: EZ-diffusion by hand

A participant has $P_c = 0.802$, $V_{RT} = 0.112$ s², $M_{RT} = 0.723$ s.

1. $L = \ln(0.802/0.198) = 1.399$.
2. $x = 1.399 \times (1.399 \times 0.643 - 1.399 \times 0.802 + 0.802 - 0.5)/0.112 = 0.998$.
3. $v = 0.1 \times 0.998^{1/4} = 0.100$.
4. $a = 0.01 \times 1.399/0.100 = 0.140$.
5. $y = -0.100 \times 0.140/0.01 = -1.399$, so MDT $= (0.140/0.200) \times (1 - e^{-1.399})/(1 + e^{-1.399}) = 0.423$ s.
6. $T_{er} = 0.723 - 0.423 = 0.300$ s.

Interpretation. Of the mean correct RT of 723 ms, about 423 ms is spent accumulating evidence and 300 ms on encoding and responding. Drift 0.10 and boundary 0.14 (in EZ units with $s = 0.1$) are typical values for a moderately difficult task.

1.1.6 Lab (2 hours)

Goals

1. Implement the GCM and the prototype model for the 5-4 structure; simulate data from the GCM.
2. Fit both models by MLE, compare them, and examine the critical A1-vs-A2 prediction.
3. Check parameter recovery for the GCM.
4. Simulate a drift-diffusion process, recover its parameters with EZ-diffusion, and interpret a 2 × 2 experiment.

Timing: Part A 20 min · Part B 25 min · Part C 15 min · Part D 40 min · Part E 20 min.

Part A: the 5-4 structure and two models (20 min)

Code · کد: `code/week14.py`, cell 1 · سلول 1

Expected output:

```
stimulus features  P(A) true  A responses / 20
      A1     1110       0.84                15
      A2     1010       0.88                18
      A3     1011       0.91                18
      A4     1101       0.60                10
      A5     0111       0.60                11
      B1     1100       0.47                10
      B2     0110       0.47                11
      B3     0001       0.16                 4
      B4     0000       0.12                 2
      T1     1001       0.58                11
      T2     1000       0.48                11
      T3     1111       0.87                20
      T4     0010       0.48                10
      T5     0101       0.26                 6
      T6     0011       0.58                12
      T7     0100       0.19                 4
```

Part B: fit and compare (25 min)

Code · کد: `code/week14.py`, cell 2 · سلول 2

Expected output:

```
GCM       NLL =  28.36  AIC =  64.72  BIC =  79.79  c = 3.82  w = [0.397
↪  0.111 0.426 0.066]
prototype NLL =  30.02  AIC =  68.04  BIC =  83.11  c = 1.90  w = [0.435
↪  0.009 0.461 0.095]
stimulus  observed   GCM  prototype
      A1      0.75 0.842      0.822
      A2      0.90 0.884      0.818
```

Interpret: 1. Which model is preferred, and how strong is the evidence? (ΔAIC $= 3.3$, so Akaike weights of about 0.84 vs. 0.16: a moderate preference, not a decisive one.) 2. Look at A1 and A2. The data show A2 (0.90) classified better than A1 (0.75). The GCM predicts this ordering (0.88 vs. 0.84); the prototype model cannot (0.82 vs. 0.82). Why does this comparison discriminate between the theories more sharply than the overall fit statistic? 3. Interpret the GCM attention weights. Which dimensions did the participant attend to, and does that make sense given the category structure?

Part C: is the GCM recoverable at this design? (15 min)

Code · کد: `code/week14.py`, cell 3 · سلول 3

Expected output (about 1 minute):

```
c: r(true, recovered) = 0.08
w1: r(true, recovered) = 0.91
w2: r(true, recovered) = 0.76
w3: r(true, recovered) = 0.89
w4: r(true, recovered) = 0.85
```

What to notice: the attention weights are recovered well (r = 0.76–0.91), but the sensitivity c is essentially not recoverable at this design (r = 0.08). With only 16 stimuli and 20 trials each, a large c and a moderate c can produce very similar response proportions (once c is large, predictions saturate), and c trades off with how concentrated the weights are. So we should interpret the attention weights from Part B but not the value $\hat{c} = 3.82$. More trials, a wider range of stimuli, or a Bayesian prior on c would help. This is exactly the kind of result a recovery study is for (Week 7).

Part D: drift-diffusion and EZ (40 min)

Code · کد: `code/week14.py`, cell 4 · سلول 4

Expected output (about 30 s):

```
instruction difficulty    Pc   MRT  v_true  v_EZ  a_true  a_EZ  Ter_EZ
      speed       easy 0.870 0.423    0.25 0.243    0.08 0.078   0.304
      speed       hard 0.732 0.466    0.12 0.115    0.08 0.088   0.288
   accuracy       easy 0.975 0.572    0.25 0.261    0.14 0.140   0.317
   accuracy       hard 0.818 0.697    0.12 0.111    0.14 0.135   0.313
```

Code · کد: `code/week14.py`, cell 5 · سلول 5

Part E: interpret the 2 × 2 experiment (20 min)

Answer in your notebook: 1. Which EZ parameter changes with the instruction manipulation, and which with difficulty? Did either manipulation affect the parameter it should *not* affect? 2. Describe how the RT distribution changes from speed to accuracy instructions (Part D figure). Which part of the distribution changes most: the leading edge or the tail? 3. A clinical study finds that patients are slower than controls but equally accurate. Sketch two different DDM accounts of this result, and say which data would distinguish them.

Exercises (increasing difficulty)

Exercise 1 ⋆. Compute $P(A \mid \text{A1})$ and $P(A \mid \text{A2})$ by hand for equal weights and $c = 2$ under both models. Do they reproduce the critical prediction?

Exercise 2 ⋆. Apply EZ to the worked example's statistics and to $P_c = 1.0$ with $N = 100$ trials (use the edge correction). What happens without the correction?

Exercise 3 ⋆⋆ (prototype data). Generate data from the prototype model and fit both models. Does the model comparison identify the generating model? Run a small model recovery (20 datasets per generating model) and build the 2×2 confusion matrix.

Exercise 4 ⋆⋆ (EZ recovery and bias). For 200 simulated participants with random (v, a, T_{er}) and only 100 trials each, compare EZ estimates with the true values. Which parameter is recovered worst? Then add across-trial drift variability to the simulator (Week 1, Exercise 5) and check which EZ estimate becomes biased.

Exercise 5 ⋆⋆⋆ (Advanced: Bayesian GCM). Fit the GCM in PyMC with priors $c \sim \text{Gamma}(2, \text{scale } 2)$ and $w \sim \text{Dirichlet}(1, 1, 1, 1)$. Compare the posterior of the attention weights with the MLE, and run a posterior predictive check on the A1-vs-A2 difference. (See Lee & Wagenmakers, chapter 17, for a hierarchical version.)

Lab check-off

Show: (1) the fit table and prediction plot for the GCM vs. the prototype model with your answer to question 2, (2) the EZ table for the 2×2 design, and (3) your answer to Part E, question 1.

1.1.7 Connection to AI & LLMs

Exemplar models in a learned "semantic space". The GCM needs a psychological space in which to measure distances. Traditionally, that space comes from similarity ratings and multidimensional scaling. Language models offer an alternative: every word or sentence is mapped to an embedding vector, and the distances between vectors are often used as a stand-in for psychological similarity. Combining embeddings with the GCM gives a model that can classify *any* new word with no extra similarity ratings. Whether embedding distances match human similarity is an empirical question, and you can test it with the tools of this course.

Code · كد: `code/week14.py`, cell 6 · سلول 6

Expected output:

```
P(fruit | tomato) = 0.902
P(fruit | knife) = 0.094
P(fruit | nutcracker) = 0.370
```

Try it (no API needed). - ⋆ Change c to 1 and to 10. How do the three predictions change, and why does "nutcracker" change the most? - ⋆⋆ Suppose 20 participants classified the three probes, with 19, 1 and 12 "fruit" responses. Fit c by MLE (binomial likelihood, Week 4) and run a G^2 test. Does the embedding space with a single parameter fit? - ⋆⋆ Discussion: the embeddings were learned from text, not from perception. For which categories might text-based similarity differ systematically from human similarity (think of color or shape)?

1.1.8 Common mistakes & misconceptions

1. Unconstrained attention weights. Without a sum-to-one constraint, c and w trade off perfectly (non-identifiable). Use a softmax reparameterization.
2. Judging exemplar vs. prototype models by overall fit only. The theoretically diagnostic predictions (A1 vs. A2) are often more informative than small differences in AIC.
3. Averaging classification data over participants who attend to different dimensions. This can make an exemplar-like individual look prototype-like (and vice versa); recall Week 5's aggregation artifacts.
4. Using EZ with incorrect units. RTs must be in seconds and the variance in s², with $s = 0.1$. Mixing ms and s gives nonsense.
5. Applying EZ when its assumptions are violated (strong bias, slow or fast errors). Check error RTs; use the full DDM if needed.
6. "A slower group has a lower drift rate." Slowness can come from a higher boundary, a lower drift, or a longer non-decision time. Only a model fitted to accuracy *and* RT distributions can tell which.
7. Forgetting the edge cases $P_c = 1$ or $P_c \leq 0.5$ in EZ.

1.1.9 Check your understanding

1. State the core difference between exemplar and prototype theories in one sentence each. Answer Exemplar: a category is represented by all stored instances, and classification depends on summed similarity to them. Prototype: a category is represented by a single summary (central tendency), and classification depends on similarity to that summary.

2. In the 5-4 structure, why is the A1-vs-A2 comparison diagnostic? Answer A1 is closer to the A prototype (3 of 4 features), so a prototype model predicts it is classified better than A2. But A2 is

very similar to other A exemplars and to no B exemplar, while A1 is similar to two B exemplars, so the exemplar model can predict A2 $\geq$ A1. The theories make opposite ordinal predictions.

3. What do the GCM parameters c and w_m represent? Answer c is sensitivity: how quickly similarity decreases with psychological distance (large c means only very close exemplars matter). w_m is the proportion of attention allocated to dimension m, which stretches distances along attended dimensions.

4. Which DDM parameter would you expect to change under (a) speed vs. accuracy instructions, (b) harder stimuli, (c) a slower motor response (e.g., foot instead of finger)? Answer (a) Boundary separation a. (b) Drift rate v. (c) Non-decision time T_{er}.

5. What three statistics does EZ-diffusion need, and what are its main assumptions? Answer Proportion correct, the variance of correct RTs, and the mean of correct RTs. It assumes an unbiased starting point ($z = a/2$), no across-trial variability in parameters, and the scaling $s = 0.1$.

6. How is the DDM related to the random walk model of Week 1? Answer It is the continuous-time, continuous-evidence limit of the random walk: as step size and step duration go to zero together, the random walk converges to a Wiener diffusion process with drift v and noise s.

1.1.10 Assignment

- A5 (MCMC from scratch) is due at the start of this week's lecture.
- No new assignment: work on the final project. If your project uses the GCM or DDM, today's code is a good starting point; cite the course materials and say what you changed.

1.1.11 Further reading

- Lee & Wagenmakers (2014), chapter 17. The Generalized Context Model, fitted in a Bayesian (and hierarchical) way to category-learning data. *(Main reading for categorization.)*
- Farrell & Lewandowsky (2018): the chapter on models of choice response time [verify chapter number] for the drift-diffusion model and its relatives.
- Verguts (2022): the chapters on categorization and on decision making [verify chapters].
- *Optional, outside the course list:* Wagenmakers, E.-J., van der Maas, H. L. J., & Grasman, R. P. P. P. (2007). An EZ-diffusion model for response time and accuracy. *Psychonomic Bulletin & Review, 14*, 3–22. Medin, D. L., & Schaffer, M. M. (1978). Context theory of classification learning. *Psychological Review, 85*, 207–238.

Preview for Week 15: learning from rewards (reinforcement learning), the mind as a Bayesian observer, and how cognitive models and AI systems inform each other.